

Invariant Estimator Framework for Autonomous Cars

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I. STATE ON $SE_2(3)$

We define the state on the group manifold $SE_2(3)$, such that we can maintain invariant observer properties while keeping the system as observable as possible:

$$X \triangleq \begin{bmatrix} {}^{\mathcal{W}}R_{\mathcal{B}} & {}^{\mathcal{W}}\mathbf{v}_{\mathcal{B}} & {}^{\mathcal{W}}\mathbf{p}_{\mathcal{B}} \\ \mathbf{0}_{1 \times 3} & 1 & 0 \\ \mathbf{0}_{1 \times 3} & 0 & 1 \end{bmatrix} \in SE_2(3) \quad (1)$$

$$\boldsymbol{\theta} = (\mathbf{b}_g, \mathbf{b}_a) \in \mathbb{R}^6$$

this has the inverse state also defined as:

$$X^{-1} = \begin{bmatrix} R^\top & -R^\top \mathbf{v} & -R^\top \mathbf{p} \\ \mathbf{0}_{1 \times 3} & 1 & 0 \\ \mathbf{0}_{1 \times 3} & 0 & 1 \end{bmatrix} \in SE_2(3) \quad (2)$$

As the state is defined on a *matrix lie group*, we also have to define the associated lie algebra state and its associated tangent vector. Both are defined below:

$$\boldsymbol{\xi}^\wedge = \begin{bmatrix} \rho_\phi^\wedge & \rho_v & \rho_p \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix} \quad (3)$$

$$\boldsymbol{\xi} = [\rho_\phi \quad \rho_v \quad \rho_p \quad \zeta_g \quad \zeta_a] \in \mathbb{R}^{15} \quad (4)$$

where the exponential on the state component of the tangent vector defined as $\exp([\boldsymbol{\xi}_{1:9}]^\wedge) = \hat{X}X^{-1} = \boldsymbol{\eta}_R$ (the right-invariant error), and $\zeta = \boldsymbol{\theta} - \boldsymbol{\theta}$.

II. PROCESS MODEL

We define the following continuous time (CT) dynamics:

$$\begin{aligned} {}^{\mathcal{W}}\dot{R}_{\mathcal{B}} &= {}^{\mathcal{W}}R_{\mathcal{B}} {}^{\mathcal{B}}\bar{\boldsymbol{\omega}}^\wedge \\ {}^{\mathcal{W}}\dot{\mathbf{v}}_{\mathcal{B}} &= {}^{\mathcal{W}}R_{\mathcal{B}} {}^{\mathcal{B}}\bar{\mathbf{a}} + {}^{\mathcal{W}}\mathbf{g} \\ {}^{\mathcal{W}}\dot{\mathbf{p}}_{\mathcal{B}} &= {}^{\mathcal{W}}\mathbf{v}_{\mathcal{B}} \\ \dot{\mathbf{b}}_g &= \mathbf{w}_{bg}, \mathbf{w}_{bg} \sim \mathcal{N}(0, \Sigma_g) \\ \dot{\mathbf{b}}_a &= \mathbf{w}_{ba}, \mathbf{w}_{ba} \sim \mathcal{N}(0, \Sigma_a) \end{aligned} \quad (5)$$

where ${}^{\mathcal{B}}\bar{\boldsymbol{\omega}}$ and ${}^{\mathcal{B}}\bar{\mathbf{a}}$ are the bias corrected inputs, which are later derived in III. And we know that these dynamics, when bias-free, are *group-affine* - that is, that they yield a log-linear error such that $\hat{\boldsymbol{\xi}} = A\boldsymbol{\xi}$ (with $\boldsymbol{\eta} = \exp(\boldsymbol{\xi}^\wedge)$) - this is further shown in the appendix. However, this is generally not the case, as shown in (CITE Barreau here), that the augmentation of biases onto the InEKF state yields an imperfect InEKF, as $\bar{\boldsymbol{\omega}}$ and $\bar{\mathbf{a}}$ both depend on $\hat{\mathbf{b}}$.

If we assume that the input measurements ${}^{\mathcal{B}}\boldsymbol{\omega}$ and ${}^{\mathcal{B}}\mathbf{a}$ are constant over some small timestep interval $[t, t+\Delta t]$ (i.e., zero-order hold), then we know that the rotation matrix propagation is defined exactly:

$${}^{\mathcal{W}}R_{\mathcal{B}}(t_k + s) = {}^{\mathcal{W}}R_{\mathcal{B},k} \exp(s[{}^{\mathcal{B}}\boldsymbol{\omega}]^\wedge) \quad (6)$$

This means that we can substitute this into the discretization of both velocity and position, with $s = u\Delta t$, for some timestep Δt . For instance, with velocity, the integrand yields:

$$\int_0^{\Delta t} \exp(s[{}^{\mathcal{B}}\boldsymbol{\omega}]^\wedge) ds = \Delta t \int_0^1 \exp(u[{}^{\mathcal{B}}\boldsymbol{\omega}]^\wedge) du = \Delta t \Gamma_1(\phi) \quad (7)$$

And the integral behaves similarly for position. This means that after discretizing, we are left with the following dynamics that obey the $SO(3)$ manifold that live on $SE_2(3)$, with $\phi \triangleq {}^{\mathcal{B}}\bar{\boldsymbol{\omega}}\Delta t$, and frames omitted from state components as in (1) for brevity:

$$\begin{aligned} R_{k+1|k} &= R_{k|k} \Gamma_0(\phi) \\ \mathbf{v}_{k+1|k} &= \mathbf{v}_{k|k} + {}^{\mathcal{W}}\mathbf{g}\Delta t + R_{k|k} \Gamma_1(\phi) \mathbf{a} \Delta t \\ \mathbf{p}_{k+1|k} &= \mathbf{p}_{k|k} + \mathbf{v}_{k|k} \Delta t + \frac{1}{2} {}^{\mathcal{W}}\mathbf{g} \Delta t^2 + R_{k|k} \Gamma_2(\phi) \mathbf{a} \Delta t^2 \\ \hat{\boldsymbol{\theta}}_{k+1|k} &= \hat{\boldsymbol{\theta}}_{k|k} \end{aligned} \quad (8)$$

III. MEASUREMENT MODEL

We formulate the measurement models in both the *right-invariant* and *left-invariant* forms (CITE this), i.e. $\mathbf{Y}_i = X^{-1}\mathbf{d}_i + \mathbf{V}_i$ for right-invariant and $\mathbf{Y}_i = X\mathbf{d}_i + \mathbf{V}_i$ for left-invariant. We define four measurement models, two in the right-invariant form (body frame centric), and two in the left-invariant form.

A. Right-Invariant Measurement Models

1) *Body-Frame Velocity Psuedo Measurement*: We define the body-frame velocity at the IMU origin as ${}^{\mathcal{B}}\mathbf{v} = {}^{\mathcal{W}}R_{\mathcal{B}}^\top {}^{\mathcal{W}}\mathbf{v}_{\mathcal{B}}$. And if we define $\mathbf{r}_w$ as the vector from the IMU origin to the center of the rear axle, by rigid body kinematics we have:

$${}^{\mathcal{B}}\mathbf{v}_{\text{axle}} = {}^{\mathcal{B}}\mathbf{v}_{\text{IMU}} + \boldsymbol{\omega}^\wedge \mathbf{r}_w \quad (9)$$

Now that we have the equation, we need to put this into Lie group measurement form. As this is body frame, we need to utilize the inverse of the state, as defined in (2). If we

define the selection vector $\mathbf{d} = (\mathbf{0}_3, 1, 0)^\top$, then we get our measurement model as:

$$X^{-1}\mathbf{d} = \begin{bmatrix} R^\top & -R^\top \mathbf{v} & -R^\top \mathbf{p} \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix} = \begin{bmatrix} -R^\top \mathbf{v} \\ 1 \\ 0 \end{bmatrix} \quad (10)$$

2) *Zero Angular Update Rate (ZARU)*: This measurement is deliberately excluded from the group, and is simply a method with which to pin $\mathbf{b}_g$ when the car is stationary, as at a zero point we know that $\tilde{\boldsymbol{\omega}} = \mathbf{b}_g + \mathbf{n}_g$. Because of this, we can directly formulate the measurement model, along with its linearized Jacobian, as the following:

$$\begin{aligned} \mathbf{z} &= \tilde{\boldsymbol{\omega}} - \hat{\mathbf{b}}_g \\ H &= \begin{bmatrix} 0 & 0 & 0 & -I_3 & 0 \end{bmatrix} \end{aligned} \quad (11)$$

B. Left-Invariant Measurement Models